

## 6.9 Other inner products

Everything so far has used the Euclidean inner product  $\mathbf{u}^T \mathbf{v}$ . That choice treats the  $n$  coordinates of observation space symmetrically: an error of a given size counts the same in every observation, and errors in different observations are unrelated. When observations have different variances or are correlated, Euclidean distance is the wrong way to measure how far  $\mathbf{y}$  is from the model space. This section shows that the whole theory carries over to any inner product, and that the result is generalized least squares.

### 6.9.1 Projection in a general inner product

Let  $\mathbf{A}$  be an  $n \times n$  symmetric positive definite matrix, and define

$$\langle \mathbf{u}, \mathbf{v} \rangle_{\mathbf{A}} = \mathbf{u}^T \mathbf{A} \mathbf{v}, \quad \|\mathbf{u}\|_{\mathbf{A}} = (\mathbf{u}^T \mathbf{A} \mathbf{u})^{1/2}.$$

This is an inner product: bilinear, symmetric, and positive on nonzero vectors. Say  $\mathbf{u}$  and  $\mathbf{v}$  are  $\mathbf{A}$ -orthogonal if  $\mathbf{u}^T \mathbf{A} \mathbf{v} = 0$ . Nothing in the proofs of Proposition 6.2.2, Theorem 6.2.3 or Theorem 6.3.1 used anything about  $\mathbf{u}^T \mathbf{v}$  except these three properties. So those results hold verbatim with  $\langle \cdot, \cdot \rangle_{\mathbf{A}}$  in place of the Euclidean inner product. In particular, every  $\mathbf{y}$  has a unique  $\mathbf{A}$ -nearest point in a subspace  $\mathcal{S}$ , characterized by  $\mathbf{A}$ -orthogonality of the error to  $\mathcal{S}$ . The map sending  $\mathbf{y}$  to it is linear, and we write it  $P_{\mathcal{S}, \mathbf{A}}$ .

#### Theorem 6.9.1

Let  $\mathbf{A}$  be positive definite and  $\mathbf{X}$  any  $n \times p$  matrix. The  $\mathbf{A}$ -orthogonal projection onto  $\mathbb{C}(\mathbf{X})$  is

$$P_{\mathbf{X}, \mathbf{A}} = \mathbf{X}(\mathbf{X}^T \mathbf{A} \mathbf{X})^{-} \mathbf{X}^T \mathbf{A},$$

for any generalized inverse. It satisfies  $P_{\mathbf{X}, \mathbf{A}}^2 = P_{\mathbf{X}, \mathbf{A}}$  and  $\mathbb{C}(P_{\mathbf{X}, \mathbf{A}}) = \mathbb{C}(\mathbf{X})$ , and the matrix  $\mathbf{A} P_{\mathbf{X}, \mathbf{A}}$  is symmetric. The matrix  $P_{\mathbf{X}, \mathbf{A}}$  itself is symmetric iff  $\mathbf{A} \mathbb{C}(\mathbf{X}) \subseteq \mathbb{C}(\mathbf{X})$ , and in general it is not.

*Proof.* Let  $\mathbf{A} = \mathbf{L} \mathbf{L}^T$  with  $\mathbf{L}$  nonsingular (for instance the Cholesky factor), and put  $\mathbf{X}_* = \mathbf{L}^T \mathbf{X}$  and  $\mathbf{y}_* = \mathbf{L}^T \mathbf{y}$ . Then  $\|\mathbf{y} - \mathbf{X} \mathbf{b}\|_{\mathbf{A}}^2 = \|\mathbf{y}_* - \mathbf{X}_* \mathbf{b}\|^2$ , so  $\mathbf{b}$  minimizes the  $\mathbf{A}$ -distance iff it is an ordinary least squares estimate for  $(\mathbf{y}_*, \mathbf{X}_*)$ . By Theorem 6.3.7 and Theorem 6.4.2, one such estimate is  $(\mathbf{X}_*^T \mathbf{X}_*)^{-} \mathbf{X}_*^T \mathbf{y}_* = (\mathbf{X}^T \mathbf{A} \mathbf{X})^{-} \mathbf{X}^T \mathbf{A} \mathbf{y}$ , and  $\mathbf{X}_* (\mathbf{X}_*^T \mathbf{X}_*)^{-} \mathbf{X}_*^T$  is invariant to the choice of generalized inverse. Multiplying on the left by  $\mathbf{L}^{-T}$  and on the right by  $\mathbf{L}^T$  shows that  $P_{\mathbf{X}, \mathbf{A}} = \mathbf{L}^{-T} \mathbf{M}_* \mathbf{L}^T$ , where  $\mathbf{M}_*$  projects onto  $\mathbb{C}(\mathbf{X}_*)$ . Idempotence and  $\mathbb{C}(P_{\mathbf{X}, \mathbf{A}}) = \mathbf{L}^{-T} \mathbb{C}(\mathbf{X}_*) = \mathbb{C}(\mathbf{X})$  follow, and  $\mathbf{A} P_{\mathbf{X}, \mathbf{A}} = \mathbf{L} \mathbf{M}_* \mathbf{L}^T$  is symmetric. For the last claim, see [Exercise \(B1\)](#). ■

The proof also gives the general principle:  *$\mathbf{A}$ -geometry is Euclidean geometry after the change of variables  $\mathbf{y} \mapsto \mathbf{L}^T \mathbf{y}$ .* Any fact about orthogonal projections has an  $\mathbf{A}$ -version, obtained by transporting it through  $\mathbf{L}$ .

Symmetry is the Euclidean signature of an orthogonal projection. The  $\mathbf{A}$ -version of Theorem 6.3.3 replaces it with self-adjointness.

### Proposition 6.9.2

An idempotent matrix  $P$  is the  $\mathbf{A}$ -orthogonal projection onto  $\mathbb{C}(P)$  iff  $\mathbf{A}P$  is symmetric, that is, iff  $\langle Pu, v \rangle_{\mathbf{A}} = \langle u, Pv \rangle_{\mathbf{A}}$  for all  $u, v$ .

*Proof.* For idempotent  $P$ , the residual  $(\mathbf{I} - P)y$  is  $\mathbf{A}$ -orthogonal to  $\mathbb{C}(P)$  for every  $y$  iff  $PTA(\mathbf{I} - P) = \mathbf{0}$ , that is, iff  $PTA = PTAP$ . If  $\mathbf{A}P$  is symmetric, then  $PTA = AP$ , and  $PTAP = APP = AP = PTA$ . Conversely,  $PTA = PTAP$  has a symmetric right-hand side, so  $PTA$  and hence its transpose  $AP$  are symmetric. ■

## 6.9.2 Generalized least squares

Suppose  $\text{Cov}(\epsilon) = \sigma^2 \mathbf{V}$  with  $\mathbf{V}$  known and positive definite. Observations with large variance should count for less, and correlated observations partly repeat each other. The natural distance is the one that makes the errors look uncorrelated:  $\|\cdot\|_{\mathbf{V}^{-1}}$ . If  $\mathbf{V} = \mathbf{L}\mathbf{L}^T$ , then  $\mathbf{L}^{-1}\epsilon$  has covariance  $\sigma^2 \mathbf{I}$ , and  $\|u\|_{\mathbf{V}^{-1}} = \|\mathbf{L}^{-1}u\|$ .

### Definition 6.9.1 — Generalized least squares

A **generalized least squares** (GLS) estimate minimizes  $(y - \mathbf{X}b)^T \mathbf{V}^{-1}(y - \mathbf{X}b)$ . Equivalently,  $\mathbf{X}\hat{\beta}_{\text{GLS}} = P_{\mathbf{X}, \mathbf{V}^{-1}}y$ , and  $\hat{\beta}_{\text{GLS}}$  solves the generalized normal equations  $\mathbf{X}^T \mathbf{V}^{-1} \mathbf{X}b = \mathbf{X}^T \mathbf{V}^{-1} y$ .

**Theorem 6.9.1** with  $\mathbf{A} = \mathbf{V}^{-1}$  is the whole of the algebra. Computationally, GLS is ordinary least squares applied to the *whitened* data  $(\mathbf{L}^{-1}y, \mathbf{L}^{-1}\mathbf{X})$ . Aitken (1935) showed that GLS gives the best linear unbiased estimator under this covariance, which is the Gauss–Markov theorem transported through  $\mathbf{L}$ . Chapter 31 proves this, and handles singular  $\mathbf{V}$  and the more realistic case where  $\mathbf{V}$  must be estimated.

When is the ordinary least squares fit already the GLS fit? Geometrically, the two projections onto  $\mathbb{C}(\mathbf{X})$  coincide iff the Euclidean and the  $\mathbf{V}^{-1}$  notions of orthogonality agree on this particular subspace.

### Theorem 6.9.3 — Kruskal

For positive definite  $\mathbf{V}$ ,  $\mathbf{M}y = P_{\mathbf{X}, \mathbf{V}^{-1}}y$  for every  $y$  iff  $\mathbb{C}(\mathbf{V}\mathbf{X}) \subseteq \mathbb{C}(\mathbf{X})$ .

*Proof.* The two projections have the same range. They agree iff they have the same null space, since an idempotent matrix is determined by its range and null space (**Proposition 6.9.4**). The null space of  $\mathbf{M}$  is  $\mathbb{C}(\mathbf{X})^\perp = \mathcal{N}(\mathbf{X}^T)$ . The null space of  $P_{\mathbf{X}, \mathbf{V}^{-1}}$  is the set of  $u$  with  $\mathbf{X}^T \mathbf{V}^{-1} u = \mathbf{0}$ , that is,  $\mathbf{V} \mathcal{N}(\mathbf{X}^T)$ . So the projections agree iff  $\mathbf{V} \mathcal{N}(\mathbf{X}^T) = \mathcal{N}(\mathbf{X}^T)$ , and since  $\mathbf{V}$  is nonsingular and the spaces have equal dimension, iff  $\mathbf{V} \mathcal{N}(\mathbf{X}^T) \subseteq \mathcal{N}(\mathbf{X}^T)$ . By symmetry of  $\mathbf{V}$ , this says  $\mathbf{X}^T \mathbf{V} u = \mathbf{0}$  whenever  $\mathbf{X}^T u = \mathbf{0}$ , that is,  $\mathbb{C}(\mathbf{V}\mathbf{X}) \perp \mathbb{C}(\mathbf{X})^\perp$ , which is  $\mathbb{C}(\mathbf{V}\mathbf{X}) \subseteq \mathbb{C}(\mathbf{X})$ .

The condition holds, for example, for equicorrelated errors  $\mathbf{V} = (1 - \rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}^T$  whenever  $\mathbf{1} \in \mathbb{C}(\mathbf{X})$  (Exercise (A1)). It also underlies the analysis of balanced split-plot designs in Chapter 33. The result is due to Kruskal (1968), and the coordinate-free argument is his.

### Example. (Autocorrelated errors)

Take eight equally spaced observations with a linear trend and error covariance  $v_{st} = 0.7^{|s-t|}$ . The  $\mathbf{V}^{-1}$ -projection onto  $\mathbb{C}(\mathbf{X})$  is idempotent and self-adjoint for  $\langle \cdot, \cdot \rangle_{\mathbf{V}^{-1}}$  but not symmetric: its largest asymmetry  $\max|p_{st} - p_{ts}|$  is 0.444. The ordinary and generalized slope estimates are 0.648 and 0.650. The listing checks each claim, including that the GLS fit is ordinary least squares on whitened data.

```
import numpy as np
rng = np.random.default_rng(9)
n = 8
X = np.column_stack([np.ones(n), np.arange(n, dtype=float)])
y = X @ np.array([1.0, 0.5]) + rng.normal(size=n)
t = np.arange(n)
V = 0.7 ** np.abs(t[:, None] - t[None, :])          # AR(1)-type covariance

Vinv = np.linalg.inv(V)
P_V = X @ np.linalg.solve(X.T @ Vinv @ X, X.T @ Vinv)  # V^{-1}-orthogonal
               projection onto C(X)

print("idempotent:", np.allclose(P_V @ P_V, P_V))
print("symmetric: ", np.allclose(P_V, P_V.T))          # False: oblique in
               the usual geometry
print("self-adjoint for <u,v> = u'V^{-1}v:", np.allclose(Vinv @ P_V, (Vinv @ P_V).T)
      )

# GLS = ordinary least squares after whitening with a square root of V^{-1}
L = np.linalg.cholesky(V)                          # V = L L'
Xw, yw = np.linalg.solve(L, X), np.linalg.solve(L, y)
beta_whitened, *_ = np.linalg.lstsq(Xw, yw, rcond=None)
beta_gls = np.linalg.solve(X.T @ Vinv @ X, X.T @ Vinv @ y)
print("GLS via P_V equals OLS on whitened data:", np.allclose(beta_gls,
      beta_whitened))
```

[Run this code online](#)

## 6.9.3 Weighted least squares

The simplest non-Euclidean geometry is a diagonal one. If the errors are uncorrelated with variances  $\sigma^2/w_i$ , then  $\mathbf{V}^{-1} = \mathbf{W} = \text{diag}(w_1, \dots, w_n)$ , and GLS minimizes  $\sum_i w_i(y_i - \mathbf{x}_{(i)}^T \mathbf{b})^2$ . Observations believed to be noisier get less weight in the distance. Whitening multiplies row  $i$  of  $\mathbf{X}$  and  $\mathbf{y}$  by  $\sqrt{w_i}$ . The fitted values are  $\mathbf{P}_{\mathbf{X}, \mathbf{W}}\mathbf{y}$ , and the natural leverages are the diagonal entries of  $\mathbf{P}_{\mathbf{X}, \mathbf{W}}$ . These equal the diagonal of the orthogonal projection onto  $\mathbb{C}(\mathbf{W}^{1/2}\mathbf{X})$  and still sum to  $\text{rank}(\mathbf{X})$  (Exercise (B3)).

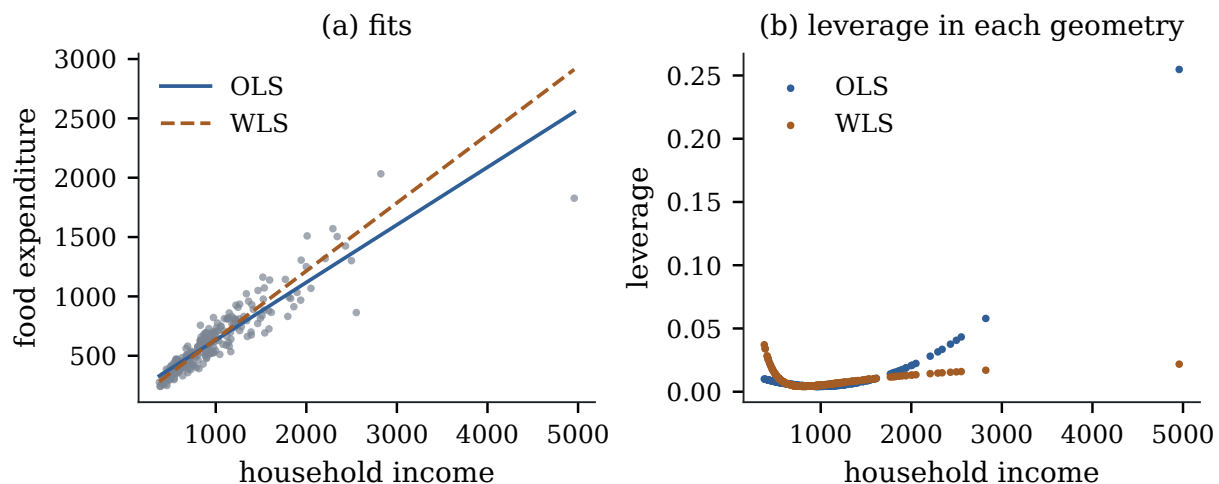


Figure 6.9.1: Engel's data. (a) Ordinary and weighted least squares fits, with weights proportional to  $1/\text{income}^2$ . (b) Leverages under each geometry. The weighted projection spreads influence away from the high-variance, high-income households.

#### Example. (Engel's food expenditure data)

Engel's 1857 survey of 235 Belgian working-class households records annual income and food expenditure. The spread of expenditure grows with income, as [Figure 6.9.1\(a\)](#) shows. Weighting by  $w_i = 1/\text{income}_i^2$  (standard deviation proportional to income) changes the fitted slope from 0.4852 to 0.5740. The change in geometry shows up more clearly in the leverages, panel (b). Under ordinary least squares the richest households have the most leverage, with a maximum of 0.255. In the weighted geometry their large variance shrinks their influence, and no household has leverage above 0.037. Whether this is the *right* geometry depends on whether the variance model holds. Chapter 21 discusses how to check it, and how to get valid standard errors when it fails.

```
import numpy as np
import statsmodels.api as sm
df = sm.datasets.engel.load_pandas().data
x, y = df["income"].to_numpy(), df["foodexp"].to_numpy()
X = np.column_stack([np.ones(len(x)), x])

w = 1.0 / x**2                                     # working model: sd proportional to
income
sw = np.sqrt(w)
beta_ols, *_ = np.linalg.lstsq(X, y, rcond=None)
beta_wls, *_ = np.linalg.lstsq(X * sw[:, None], y * sw, rcond=None) # OLS on (W
~1/2 X, W~1/2 y)

# leverages in the weighted geometry: squared row norms of an orthonormal basis of C
(W~1/2 X)
Qw, _ = np.linalg.qr(X * sw[:, None])
h_wls = np.sum(Qw**2, axis=1)
Q, _ = np.linalg.qr(X)
h_ols = np.sum(Q**2, axis=1)
print("OLS:", beta_ols, " WLS:", beta_wls)
print("max leverage OLS %.3f, WLS %.3f" % (h_ols.max(), h_wls.max()))
```

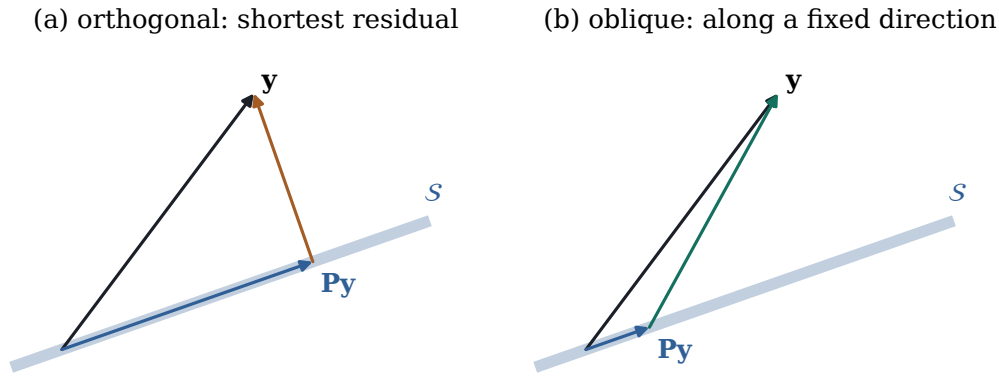


Figure 6.9.2: Projections onto a line  $\mathcal{S}$  in  $\mathbb{R}^2$ . (a) The orthogonal projection: the residual is perpendicular to  $\mathcal{S}$ , and  $\mathbf{P}\mathbf{y}$  is the nearest point. (b) An oblique projection moves  $\mathbf{y}$  along a fixed direction that is not perpendicular to  $\mathcal{S}$ . The result is idempotent but is not the nearest point, and the residual is longer.

[Run this code online](#)

## 6.9.4 Oblique projections

The matrices  $\mathbf{P}_{\mathbf{X},\mathbf{A}}$  are idempotent without being symmetric. In the Euclidean geometry of  $\mathbb{R}^n$  they are **oblique projections**. Idempotent matrices in general have a clean description.

### Proposition 6.9.4

Let  $\mathbf{P}$  be idempotent. Then  $\mathbb{R}^n = \mathbb{C}(\mathbf{P}) \oplus \mathcal{N}(\mathbf{P})$ , and  $\mathbf{P}$  maps each  $\mathbf{y} = \mathbf{u} + \mathbf{v}$  ( $\mathbf{u} \in \mathbb{C}(\mathbf{P})$ ,  $\mathbf{v} \in \mathcal{N}(\mathbf{P})$ ) to  $\mathbf{u}$ . Conversely, for any decomposition  $\mathbb{R}^n = \mathcal{S} \oplus \mathcal{T}$  there is exactly one idempotent  $\mathbf{P}$  with  $\mathbb{C}(\mathbf{P}) = \mathcal{S}$  and  $\mathcal{N}(\mathbf{P}) = \mathcal{T}$ , the *projection onto  $\mathcal{S}$  along  $\mathcal{T}$* . It is an orthogonal projection iff  $\mathcal{T} = \mathcal{S}^\perp$ , that is, iff  $\mathbf{P}$  is symmetric.

*Proof.* Write  $\mathbf{y} = \mathbf{P}\mathbf{y} + (\mathbf{I} - \mathbf{P})\mathbf{y}$ . The first term lies in  $\mathbb{C}(\mathbf{P})$ , and  $\mathbf{P}(\mathbf{I} - \mathbf{P})\mathbf{y} = \mathbf{0}$  puts the second in  $\mathcal{N}(\mathbf{P})$ . If  $\mathbf{u} = \mathbf{P}\mathbf{w} \in \mathcal{N}(\mathbf{P})$  then  $\mathbf{u} = \mathbf{P}^2\mathbf{w} = \mathbf{P}\mathbf{u} = \mathbf{0}$ , so the sum is direct. For  $\mathbf{u} \in \mathbb{C}(\mathbf{P})$ ,  $\mathbf{P}\mathbf{u} = \mathbf{u}$  (write  $\mathbf{u} = \mathbf{P}\mathbf{w}$ ), so  $\mathbf{P}(\mathbf{u} + \mathbf{v}) = \mathbf{u}$ . Conversely, given  $\mathcal{S} \oplus \mathcal{T}$ , the map  $\mathbf{u} + \mathbf{v} \mapsto \mathbf{u}$  is well defined and linear by uniqueness of the decomposition, and it is idempotent with the required range and null space. It is unique because its values are forced on all of  $\mathbb{R}^n$ . If  $\mathcal{T} = \mathcal{S}^\perp$ , the map is the orthogonal projection, which is symmetric. If  $\mathbf{P}$  is symmetric, then  $\mathcal{N}(\mathbf{P}) = \mathcal{N}(\mathbf{P}\mathbf{T}) = \mathbb{C}(\mathbf{P})^\perp$  by Lemma 6.2.1. ■

Figure 6.9.2 shows the difference. The nearest-point property belongs to the orthogonal projection alone. An oblique projection can move a vector *farther* away: its operator norm exceeds one unless it is orthogonal ([Exercise \(B2\)](#)). The GLS projection is still a nearest-point map, but in the  $\mathbf{V}^{-1}$  norm, not the Euclidean one. Which geometry is “right” is not a mathematical question. It is decided by the covariance of the errors.

## 6.9.5 Exercises

### 6.9.5.1 A. Check your understanding

#### Exercise A1.

Let  $\mathbf{V} = (1 - \rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}^T$  with  $-1/(n-1) < \rho < 1$ . Show that  $\mathbf{V}$  is positive definite and that  $\mathbb{C}(\mathbf{V}\mathbf{X}) \subseteq \mathbb{C}(\mathbf{X})$  whenever  $\mathbf{1} \in \mathbb{C}(\mathbf{X})$ . Conclude that ordinary least squares is generalized least squares for equicorrelated errors in any model with an intercept.

*Solution.*  $\mathbf{V}\mathbf{1} = (1 + (n-1)\rho)\mathbf{1}$  and  $\mathbf{V}\mathbf{u} = (1 - \rho)\mathbf{u}$  for  $\mathbf{u} \perp \mathbf{1}$ . So the eigenvalues are  $1 + (n-1)\rho$  and  $1 - \rho$ , both positive exactly when  $-1/(n-1) < \rho < 1$ . Then  $\mathbf{V}\mathbf{X} = (1 - \rho)\mathbf{X} + \rho\mathbf{1}(\mathbf{1}^T\mathbf{X})$ , whose columns lie in  $\mathbb{C}(\mathbf{X})$  when  $\mathbf{1}$  does. [Theorem 6.9.3](#) applies.

### 6.9.5.2 B. Practice

#### Exercise B1.

Complete the proof of [Theorem 6.9.1](#). Show that  $\mathbf{P}_{\mathbf{X},\mathbf{A}}$  is symmetric iff it equals  $\mathbf{M}$ , and deduce from [Theorem 6.9.3](#) (with  $\mathbf{V} = \mathbf{A}^{-1}$ ) that this happens iff  $\mathbf{A}\mathbb{C}(\mathbf{X}) \subseteq \mathbb{C}(\mathbf{X})$ .

#### Exercise B2.

Let  $\mathbf{P} \neq \mathbf{0}$  be idempotent. Show that the operator norm  $\|\mathbf{P}\|_2 = \max_{\|\mathbf{u}\|=1} \|\mathbf{P}\mathbf{u}\|$  is at least 1, with equality iff  $\mathbf{P}$  is symmetric. *Hint:* if  $\mathbf{P}$  is not symmetric, find  $\mathbf{u} \in \mathbb{C}(\mathbf{P})$  and  $\mathbf{v} \in \mathcal{N}(\mathbf{P})$  with  $\mathbf{u}^T\mathbf{v} \neq 0$ , and consider  $\mathbf{u} + t\mathbf{v}$  for small  $t$ .

*Solution.* For  $\mathbf{u} \in \mathbb{C}(\mathbf{P})$ ,  $\mathbf{P}\mathbf{u} = \mathbf{u}$ , so  $\|\mathbf{P}\|_2 \geq 1$ . If  $\mathbf{P}$  is symmetric,  $\|\mathbf{P}\mathbf{u}\| \leq \|\mathbf{u}\|$  by Proposition 6.3.4(d). Suppose  $\mathbf{P}$  is not symmetric. Then  $\mathcal{N}(\mathbf{P}) \neq \mathbb{C}(\mathbf{P})^\perp$  by [Proposition 6.9.4](#). The two spaces have the same dimension, so some  $\mathbf{v} \in \mathcal{N}(\mathbf{P})$  is not orthogonal to  $\mathbb{C}(\mathbf{P})$ . Choose a unit  $\mathbf{u} \in \mathbb{C}(\mathbf{P})$  with  $\mathbf{u}^T\mathbf{v} > 0$ . For  $t > 0$ ,  $\mathbf{P}(\mathbf{u} - t\mathbf{v}) = \mathbf{u}$ , while  $\|\mathbf{u} - t\mathbf{v}\|^2 = 1 - 2t\mathbf{u}^T\mathbf{v} + t^2\|\mathbf{v}\|^2 < 1$  for small  $t$ . So  $\mathbf{P}$  stretches  $\mathbf{u} - t\mathbf{v}$ , and  $\|\mathbf{P}\|_2 > 1$ .

#### Exercise B3.

For weighted least squares,  $\mathbf{V} = \text{diag}(1/w_1, \dots, 1/w_n)$  with  $w_i > 0$ . Show that the diagonal entries of  $\mathbf{P}_{\mathbf{X},\mathbf{V}^{-1}}$  equal those of the orthogonal projection onto  $\mathbb{C}(\mathbf{W}^{1/2}\mathbf{X})$ , where  $\mathbf{W} = \text{diag}(w_i)$ , and that they sum to  $\text{rank}(\mathbf{X})$ . Interpret them as leverages.